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RESEARCH ARTICLE

Detecting Narcissistic Personality Disorder (NPD): A Hybrid Regex and NLP-Based AI Approach With Phase-Aware Classification

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ABSTRACT Narcissistic Personality Disorder (NPD) is considered one of the three malevolent personality traits comprising the 'Dark Triad' alongside Machiavellianism and Psychopathy. Recent advances in computational psycholinguistics have demonstrated the potential of natural language processing (NLP) for the detection of personality disorders. To address the complexities of detecting nuanced linguistic patterns associated with NPD and abuse cycles, hybrid models that integrate rule-based and deep learning approaches have been proposed. Our approach synergises a transparent Regex based system for explicit markers with a fine-tuned, domain-adapted BERT model for implicit, contextual patterns. Crucially, we validated this hybrid system through a rigorous three-stage process, demonstrating a replicable methodology that successfully bridges the domain gap to real-world proxy data for toxic online discourse. This work provides a robust foundation for developing computational tools to aid researchers and clinicians in analysing textual data for patterns relevant to narcissistic dynamics.

INDEX TERMS Narcissistic personality disorder, machine learning, natural language processing, computational psycholinguistics, domain adaptation, synthetic data generation, mental health AI, responsible AI, explainable AI, abuse cycle detection, hybrid classification.

I. INTRODUCTION

The proliferation of social media platforms has transformed human interaction, providing an unprecedented volume of unstructured data that reflects individual behaviours. Narcissistic Personality Disorder (NPD) is a core component of the "Dark Triad"(Fig. 1), a cluster of three malevolent personality traits that also includes Machiavellianism and Psychopathy. These traits are characterised by socially aversive qualities, often involving a manipulative and callous interpersonal style [1].

NPD presents a significant challenge in psychological assessment and interpersonal relationships due to its complex nature. Characterised by a constellation of traits, NPD profoundly impacts individuals and those around them. These traits include overt behaviours like grandiosity, need for admiration, arrogance and entitlement as well as insidious patterns such as lack of empathy, envy, exploitation and manipulation [2]. Individuals with NPD often display preoccupations with fantasies of success and power [3], engage in controlling behaviours [4] and employ defence mechanisms like projection and gaslighting [5]. These tendencies can escalate into abusive behaviours—emotional, verbal, physical, financial [6] or even cyber abuse [7]—causing substantial distress to those involved. NPD traits can be classified into three domains: Antagonism (e.g., grandiosity, manipulativeness), Negative Affectivity (e.g., emotional lability, hostility) and Disinhibition (e.g., impulsivity, irresponsibility) [8].

The narcissistic abuse cycle unfolds in distinct phases: idealisation, devaluation, discard and hoovering [9] as shown in Fig. 2. During idealisation, the abuser creates a false sense of security through excessive praise [10]. This shifts to devaluation, marked by criticism and emotional manipulation that erodes the victim's self-esteem [9]. The cycle culminates in the rejection phase with abrupt relationship termination,

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FIGURE 1. The dark triad. (Source: Authors' own elaboration).

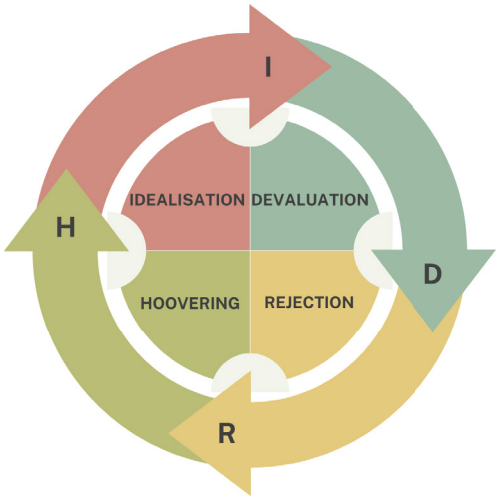

FIGURE 2. Narcissistic abuse cycle. (Source: Authors' own elaboration).

leaving victims in emotional turmoil [9]. Hoovering often follows as the abuser attempts to re-establish control through manipulative tactics [11]. Current clinical assessments relying on self-reports and behavioural observations are inherently limited by subjectivity and recall bias, while existing computational approaches often fail to capture the dynamic evolution of abusive language across different stages of the abuse cycle [11].

Building upon foundational work in sentiment analysis [12] and affective computing [13], Our research addresses these challenges by introducing a novel hybrid framework. This framework is designed to leverage the transparency of rule-based pattern matching for overt NPD markers while harnessing the contextual understanding of a fine-tuned BERT model to identify more subtle and implicit forms of abusive language. A phase-aware, multi-label classification system that maps linguistic features to relevant abuse cycle stages and any of 24 co-occurring NPD traits. This approach addresses critical gaps in current research by leveraging both psycholinguistic insights and advanced machine learning techniques.

To support this framework, we developed a dataset of over 5000 examples covering 24 NPD traits across all abuse cycle phases. While not clinically validated, the dataset exhibits strong linguistic parallels to established clinical corpora such as the Narcissistic Personality Inventory (NPI) [2] discourse dataset for grandiosity patterns and the Psychological Abuse in Relationships Scale (PARS) linguistic corpus for devaluation markers. The synthetic generation process employed clinically-informed templates to ensure ecological validity.

Our framework achieves its strong performance through technical innovations that enhance both accuracy and interpretability. By employing a prioritised classification cascade, we first leverage a high-precision regex system to detect explicit markers. If no explicit marker is found, the system then utilises a domain-adapted deep learning model for nuanced contextual analysis. This synergistic logic results in a highly interpretable model that achieves a final F1-score of 0.82 on real-world data. Furthermore, to address the challenge of analysing extended texts, we implemented chunked text processing [14], which enables the efficient detection of abuse patterns in long-form content. Together, these advancements create a scalable and reliable methodology for computational psycholinguistics.

A. RESEARCH IMPLICATIONS

The implications of this research are interdisciplinary with significant potential applications in therapeutic settings, legal contexts and digital safety. The framework aligns with recent IEEE standards for explainable AI in mental health applications (IEEE P7003) [15] while addressing limitations identified in static trait analysis in NPD research. By integrating phase-aware approaches with state-of-the-art NLP techniques, this study contributes significantly to advancing computational psycholinguistics.

B. RESEARCH CONTRIBUTIONS

This work makes three primary contributions:

1) A validated end-to-end framework for real-world generalisation in data-scarce domains. Our methodology bridges the synthetic-to-real domain gap by integrating procedural data generation, Domain-Adaptive Pre-training (DAPT) and a rigorous three-stage validation process.
2) An empirically validated hybrid architecture that synergistically combines a high-precision regex classifier for explicit linguistic markers with a high-recall, domain-adapted BERT model. The NLP component uses a multi-label framework to capture the co-occurrence of nuanced psychological traits.
3) A phase-aware classification system for the narcissistic abuse cycle. Moving beyond static trait analysis, this system provides a dynamic and context-aware tool by mapping linguistic features to the distinct stages of abusive communication (idealisation, devaluation, discard and hoovering).

The significance of this research lies in its interdisciplinary applications, offering a scalable tool for psychological

assessment, a proactive mechanism for platform moderation and a data-driven approach for criminal justice profiling.

This introduction (Section I) outlines the motivation, reviews the state-of-the-art and positions our contribution within the evolving landscape of computational psychology. Section II reviews the relevant literature and identifies key research gaps. Section III details our complete methodology, including the data curation strategy and the hybrid model architecture. Section IV presents the results of our multi-stage empirical evaluation. Finally, Section V discusses the implications and limitations of our findings and proposes directions for future work.

II. LITERATURE REVIEW

The rise of social media offers vast textual data for analysing psychological traits like Narcissistic Personality Disorder (NPD). NPD involves traits like grandiosity and lack of empathy, as outlined in the DSM-5 [3]. The narcissistic abuse cycle, with stages like idealisation and devaluation, is crucial for understanding its impact. Detecting these in text, especially from social media, can aid clinical, legal and digital safety applications. Computational psycholinguistics, employing Natural Language Processing (NLP), presents a promising avenue for detection by analysing language patterns in text.

A. COMPUTATIONAL APPROACHES TO PERSONALITY AND MENTAL HEALTH DETECTION

Recent advancements in NLP have enabled the detection of personality traits and mental health disorders from textual data, leveraging social media and other online platforms. National Library of Medicine [13] utilised machine learning and NLP to build an open-vocabulary language model from Facebook posts, successfully predicting personality traits based on the Five Factor Model (FFM) [16]. Similarly, the Massachusetts Institute of Technology [17] developed an automated depression-detection algorithm by analysing interview transcripts, demonstrating the potential of NLP in mental health applications. Other studies have extended this to detecting anxiety, PTSD and suicidal tendencies, often using social media data [18], [19]. However, these efforts predominantly focus on broader mental health issues like depression (45%) and suicide (20%) with less attention to personality disorders, particularly NPD [20].

The literature reveals a gap in computational research specifically targeting NPD. While general personality detection, such as through the Big Five traits (openness, conscientiousness, extraversion, agreeableness and neuroticism), has been explored [19], studies like those by Jayaratne and Jayatilleke [21] examine personality traits in relation to job performance rather than clinical disorders [22].

B. HYBRID MODELS IN NLP FOR ENHANCED DETECTION

Two models combine the high precision of rule-based systems, such as regular expressions for explicit NPD markers, with the contextual understanding of deep learning models like BERT for implicit abuse patterns. A review by [27] highlights their transformative potential across NLP tasks, including text classification and sentiment analysis by leveraging complementary strengths. For instance, hybrid approach combining machine learning and a rule-based expert system [28] demonstrated improved accuracy in text categorisation by fusing rule-based patterns with machine learning, a methodology adaptable to NPD detection. The advantages of hybrid models include enhanced interpretability, crucial for clinical applications and robustness in handling diverse linguistic variations. However, challenges such as computational overhead and model interpretation complexities are noted, particularly in balancing performance and explainability [29].

C. SYNTHETIC DATA IN MENTAL HEALTH NLP

The development of NLP models for sensitive domains like mental health faces significant challenges due to the limited availability of large-scale, ethically sourced datasets. Existing datasets often lack specific labels for NPD traits and abuse cycle stages and privacy risks are high due to the personal nature of the content [30]. To address this, synthetic data generation has been proposed as a solution. Iveson et al. presented an approach to generate artificial clinical documents for mental health, applying it to discharge summaries from mental healthcare providers [31]. Their work included extensive evaluations, such as text preservation measures and clinical validity assessments, demonstrating the potential of synthetic data in downstream NLP tasks like text classification. Synthetic data ensures privacy and controlled label distributions, but challenges include generalisation to real-world scenarios and potential biases from rule-based templates.

The study plans future enhancements incorporating domain adaptation techniques to bridge this gap, aligning with findings from leveraging the potential of synthetic text for AI in mental healthcare -frontiers [32]. While synthetic data ensures privacy and controlled label distributions, its primary limitation is the synthetic-to-real domain gap. Bridging this gap through techniques like domain adaptation, which our study explores, remains a critical challenge in the field.

D. DETECTING ABUSIVE LANGUAGE AND RELEVANCE TO NPD

While there is extensive research on detecting general abusive language, hate speech and offensive content using NLP, these studies often focus on broader categories like sexism and racism rather than the specific patterns of narcissistic abuse [33]. For example, Nobata et al. developed a machine learning method to detect hate speech, achieving high F1 scores, but this work does not address the cyclical nature or manipulative tactics unique to NPD [33].

TABLE 1. Summary of background literature and research gaps. (Source: Authors' own elaboration).

No.	Title	Importance to Study	Gap Identified
[1]	Sentiment Analysis of Social Media Data Using Machine Learning Techniques [23]	Provides a machine learning framework for sentiment analysis in social media, adaptable to NPD traits.	Does not target NPD or personality disorders; focuses on general sentiment.
[2]	Deep Learning Based Sentiment Analysis for Social Media Data [24]	Enhances sentiment detection with deep learning, applicable to nuanced NPD linguistic cues.	lacks specificity to NPD or personality analysis; broad social media focus.
[3]	Emotion Recognition in Conversations Using Machine Learning Techniques [16]	Offers emotion analysis techniques, foundational for detecting NPD emotional cycles.	Not specific to social media or NPD and focuses on general conversations.
[4]	A Machine Learning Approach for Personality Traits Detection in Social Media [19]	Directly relevant for personality detection in social media, a stepping stone to NPD.	Broad personality traits (e.g., Big Five), not NPD-specific or cyclic patterns.
[5]	Machine learning techniques for emotion detection and sentiment analysis: current state, challenges and future directions [18]	Links emotion/sentiment to criminal justice, relevant for NPD's manipulative traits.	Focuses on criminal investigation, not NPD or social media chats specifically.
[6]	A Survey on Sentiment Analysis and Opinion Mining: Techniques and Applications [25]	Comprehensive overview of sentiment analysis methods, informs methodology design.	No focus on NPD, personality, or specific applications like clinical or justice contexts.
[7]	The Future of AI in Detecting Narcissistic Personality [26]	Highlights societal need for NPD detection in digital spaces, aligns with study's implications.	Lacks technical rigour or empirical evidence; conceptual rather than computational.
[8]	The Development and Psychometric Properties of LIWC2015 [12]	Validates linguistic markers for personality, widely used in psychology and adaptable to NPD.	Static tool, not optimised for real-time social media or NPD cycles.

This review identifies several key gaps in the existing literature, which are detailed in Table 1 and summarised as follows:

1) A lack of NPD-specific computational research compared to other mental health issues
2) Limited focus on the dynamic, cyclical nature of narcissistic abuse
3) Challenges in data accessibility for training NLP models in sensitive domains
4) The need for interpretable and robust detection methods.
5) The absence of a proven, end-to-end framework for validating models trained on synthetic data against real-world text and a lack of methodologies to systematically mitigate the performance degradation caused by the synthetic-to-real domain gap.

III. MATERIALS AND METHODS

This section details the methodology employed to develop and evaluate a hybrid system for detecting Narcissistic Personality Disorder (NPD) traits and identifying narcissistic abuse cycle stages in textual data. The approach combines rule-based pattern matching using Regular Expressions (Regex) with deep learning-based Natural Language Processing (NLP) using fine-tuned hybrid domain adapted BERT models.

A. DATASET CURATION AND PREPARATION

The development of robust Natural Language Processing (NLP) models for sensitive domains like Narcissistic Personality Disorder (NPD) is constrained by the scarcity of large-scale, ethically sourced and specifically annotated public datasets. To overcome these limitations, our study employed a two-part data strategy: a data generation for initial supervised training and a collection of authentic real-world texts for unsupervised adaptation and final validation.

1) PURPOSE-BUILT DATASET FOR SUPERVISED TRAINING

To establish a controlled training environment and ensure comprehensive coverage, a dataset of over 5,000 examples were procedurally generated.

a: TAXONOMY AND KEYWORD RESOURCES

We first defined a comprehensive taxonomy based on the DSM-5 and associated clinical literature, consisting of 24 NPD traits (Table 2) and 4 canonical abuse cycle stages (Table 3). For the rule-based component of our model, an extensive library of regular expressions was developed for each category. As shown in Fig. 4 and Fig. 3, a substantial number of keywords were curated for each trait (approx. 450-500) and stage (approx. 560-600), demonstrating a

TABLE 2. Mapping of trait IDs to narcissistic traits. (Source: Authors' own elaboration).

Trait ID	Narcissistic Trait
T1	Grandiosity [8]
T2	Need for Admiration [8]
T3	Lack of Empathy [8]
T4	Envy [8]
T5	Arrogance [8]
T6	Exploitation [8]
T7	Sense of Entitlement [8]
T8	Manipulation [34]
T9	Belief in Specialness [8]
T10	Preoccupation with Fantasies [8]
T11	Difficulty Handling Criticism [34]
T12	Fantasies of Success [8]
T13	Perfectionism [34]
T14	Anxiety [34]
T15	Gaslighting [34]
T16	Intense Anger/Rage [8]
T17	No Boundaries [34]
T18	Controlling [34]
T19	Fear [34]
T20	Lack of Accountability [34]
T21	Verbal Abuse [34]
T22	Difficulty Forming Relationships [34]
T23	Emotional Instability [34]
T24	Intimidation [34]

TABLE 3. Mapping of stage IDs to narcissistic abuse cycle stages. (Source: Authors' own elaboration).

Stage ID	Abuse Cycle Stage
A1	Idealisation
A2	Devaluation
A3	Discard
A4	Hoovering

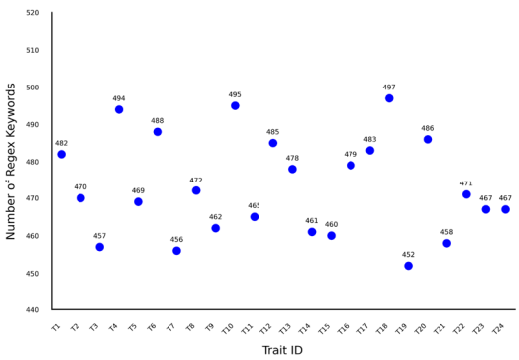

FIGURE 3. Number of regex keywords per narcissistic trait ID. (Source: Authors' own elaboration).

comparable level of linguistic detail across the taxonomy. The thematic distribution of these keywords is further detailed in Fig. 5.

b: DATA GENERATION AND COMPOSITION

Using the defined taxonomy, text samples were generated via a rule-based model. For each of the 24 NPD traits, an average of 15-20 base templates were created to capture its linguistic expression within each of the four abuse cycle stages. For every trait-stage combination ($t_i \in T$, $a_j \in A$), these structured f-string templates were utilised to embed characteristic keywords, phrases and sentiment patterns.

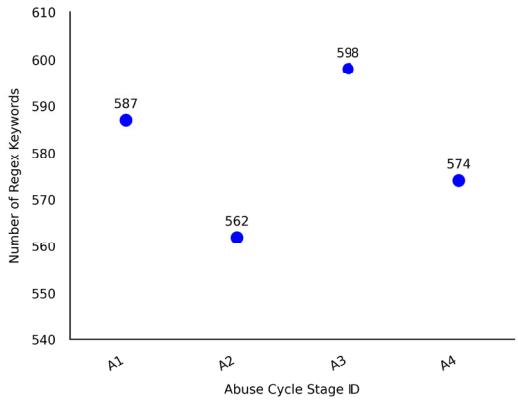

FIGURE 4. Number of regex keywords per abuse cycle stage ID. (Source: Authors' own elaboration).

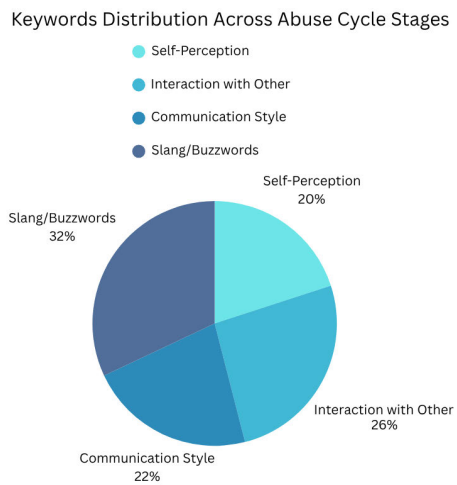

FIGURE 5. Keywords distribution across abuse cycle stages. (Source: Authors' own elaboration).

To enhance linguistic diversity and mitigate template bias, several variability techniques were implemented. For example, a single base template for 'Entitlement' such as "I deserve something because of who I am" could be varied through synonym substitution ("I am owed something; I'm more important") and restructuring ("Considering my status, it is only right that I get something"). This process, combined with randomised pronoun selection and multiple rewording strategies, resulted in a dataset, D, containing over 5,000 labelled samples, where each sample is a tuple (text_k, t_k, s_k). The dataset features a balanced distribution, with each abuse cycle stage constituting roughly 25% of the samples. Finally, this dataset was partitioned into training (80%) and testing (20%) subsets using stratified sampling.

2) AUTHENTIC DATASETS FOR ADAPTATION AND VALIDATION

a: CORPORA SELECTION AND PURPOSE

To ensure the model's real-world generalisability, a collection of authentic texts was curated for two distinct purposes. The unlabelled text from the Hate Speech and Offensive Language

Dataset [35] was used as the corpus for the unsupervised Domain-Adaptive Pre-training (DAPT) phase. For final, multi-stage validation, we used three datasets representing diverse forms of online discourse:

- Myers-Briggs Personality Type (MBTI) Dataset [36]
- PANDORA Talks Dataset [37]
- Jigsaw Toxic Comment Classification Challenge dataset [38]

The real-world relevance of our approach was further benchmarked against data from platforms like the Administrative Data Research UK (ADR UK) [39] and Harmony Data [40].

b: DATA PRE-PROCESSING

The text from these external real-world datasets consists of unstructured, multi-sentence posts. To prepare this data for our sentence-level analysis pipeline, we implemented a pre-processing step. Each post was first cleaned to remove URLs and non-ASCII characters. Subsequently, the cleaned text was segmented into individual sentences using the `sent_tokenize` function from the Natural Language Toolkit (NLTK) library. Each resulting sentence was then processed independently by our hybrid classification system.

B. HYBRID CLASSIFICATION MODEL ARCHITECTURE

Our framework is built upon a hybrid architecture that integrates two distinct but complementary components: a high-precision, rule-based regex classifier and a high-recall, context-aware NLP classifier. This section details the design of each component individually before the subsequent section explains their synergistic integration and decision logic, which is visualised in Fig. 6 and detailed in algorithm 1.

1) REGEX CLASSIFIER

The rule-based classification component employs a Deterministic Finite Automaton (DFA) approach, actualised through a library of curated regular expressions. This sub-system is engineered to execute high-precision lexical pattern matching against input text using a curated library of regular expressions compiled from over 500 keywords and phrasal templates. To enhance robustness and handle informal communication, this library was significantly augmented with over 200 relevant slang terms for each category. Two primary sets of regular expressions are utilised:

- `npd_traits_regex`: Captures explicit references to NPD-related behaviours and expressions.
- `abuse_cycle_regex`: Identifies key terms and structured phrases mapping to different stages of the abuse cycle.

If the regex engine finds a match, the system immediately assigns the corresponding label with a maximal confidence score (C=1.0). This component thus acts as a highly reliable and interpretable filter, ensuring that unambiguous patterns are classified before the text is passed to the more nuanced NLP model.

2) NLP CLASSIFIER (DOMAIN-ADAPTED AND FINE-TUNED BERT)

The NLP classifier is designed to understand nuanced and implicit language that lacks explicit keywords. Its training and architecture were developed to handle the complexities of real-world text and co-occurring psychological traits.

a: DOMAIN ADAPTATION AND SUPERVISED FINE-TUNING

Unlike the regex classifier, the NLP classifier learns through a two-stage process that combines broad linguistic knowledge with specific task-based training. The first stage is unsupervised domain adaptation, where the base BERT model is pre-trained on a large, unlabelled real-world corpus (the Hate Speech and Offensive Language Dataset [35]) to learn the vocabulary and nuances of authentic online communication. In the second stage, supervised fine-tuning, this newly adapted model is then trained on the full text of the 5,000+ labelled synthetic sentences from our training corpus. This second stage teaches the domain-aware model the specific task of identifying NPD traits and abuse cycle stages.

b: MULTI-LABEL CLASSIFICATION FOR CO-OCCURRING TRAITS

Recognising that narcissistic communication often involves multiple manipulative tactics simultaneously, the NLP classifiers were designed using a multi-label classification framework. This approach allows the model to assign a set of zero, one, or multiple labels to a given sentence, better reflecting the complexity of real-world abusive language where traits such as Gaslighting and Grandiosity can co-occur. The multi-label architecture is implemented through three key design features:

- The output layer utilises a Sigmoid activation function, which computes an independent probability for each potential label.
- The model is optimised using a Binary Cross-Entropy loss function, which evaluates the prediction for each label as an independent binary decision.
- During inference, a probability threshold of 0.5 is applied to the output of the Sigmoid layer to generate the final set of predictions.

This multi-label approach was applied to both the NPD trait and the abuse cycle stage classifiers. While it is more common for a sentence to reflect a single abuse stage, this framework allows the model to detect ambiguous or transitional statements that may contain elements of multiple phases (e.g., both devaluation and hoovering), reflecting the complexity of real-world interactions.

3) CHUNKING MECHANISM

To handle input texts exceeding BERT's 512-token limit, a chunking strategy was implemented. The text is segmented

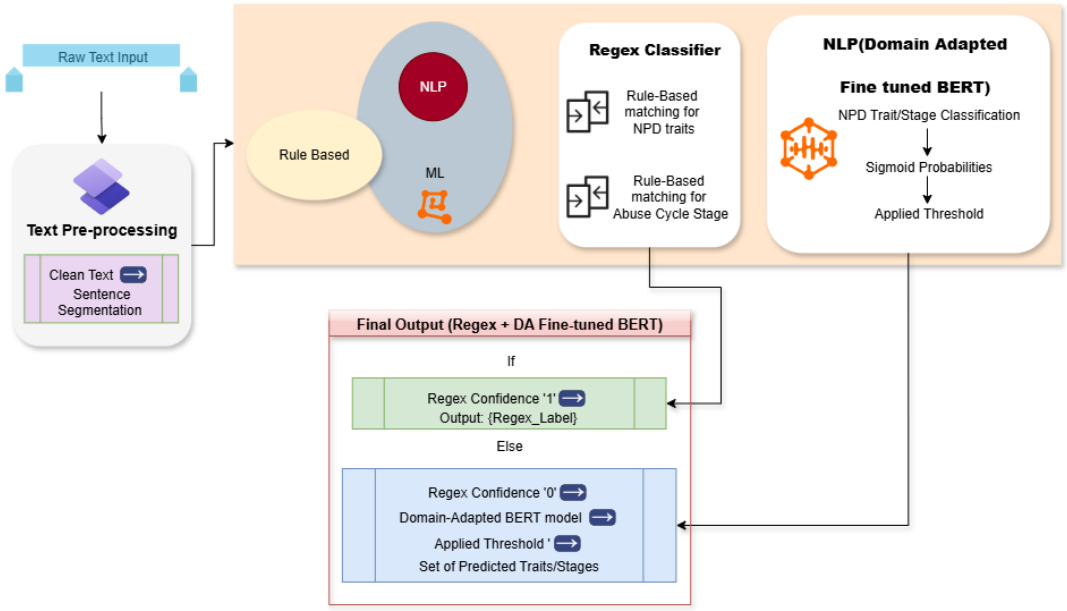

FIGURE 6. Pipeline for NPD trait and abuse-stage detection. (Source: Authors' own elaboration).

into overlapping chunks and the NLP classifier generates an independent probability vector (containing a probability for each label) for each chunk. To produce a single output for the entire text, these vectors are aggregated by taking the element-wise maximum across all chunks. This results in a final probability vector that represents the highest confidence score for each potential label observed anywhere in the text. The standard prediction threshold of 0.5 is then applied to this final vector to generate the set of predicted labels for multi-label classification, with threshold tuning noted as a future optimisation.

C. SYNERGISTIC INTEGRATION AND DECISION LOGIC

The system's novelty lies in its synergistic integration of a rule-based classifier and a deep learning model, which operate in a sequential cascade to maximise precision while ensuring comprehensive semantic coverage, as detailed in algorithm 1.

This process prioritises transparent, high-certainty classifications. Input text is first analysed by the high-precision regex classifier to identify any unambiguous, explicit markers of NPD traits or abuse cycle stages. If a pattern is matched, the system makes an immediate classification with maximum confidence (1.0), acting as a highly interpretable first-pass filter. If and only if no explicit marker is found, the text is passed to the fine-tuned, domain-adapted BERT model. This model performs a deep semantic analysis to classify subtle or implicit language that would otherwise be missed. This prioritised cascade creates a system more robust than either technique alone by balancing the precision of symbolic rules with the recall of deep learning. For complete transparency, the final output includes the predicted labels and specifies which method (Regex or NLP) was used to make the prediction.

IV. RESULTS

The empirical evaluation of the hybrid classification system was conducted through a structured, multi-stage analysis designed to transparently assess its performance and generalisability. The primary objective of this evaluation was threefold:

- Stage 1(IV-A) to establish the model's baseline efficacy on the controlled dataset;
- Stage 2(IV-B) to empirically quantify the performance degradation attributable to the synthetic-to-real domain gap when tested against diverse, real-world text corpora;
- Stage 3(IV-C) to demonstrate the successful mitigation of this gap through the application of Domain-Adaptive Pre-training (DAPT), thereby validating the robustness of the final, adapted model.

This staged approach provides a comprehensive narrative of the model's development into a system suitable for real-world application.

A. STAGE 1: BASELINE MODEL PERFORMANCE ON SYNTHETIC DATA

To establish a performance baseline, two standard BERT models were fine-tuned on a synthetically generated dataset for the distinct tasks of Narcissistic Personality Disorder (NPD) trait classification and abuse cycle stage classification. The models were trained for 10 epochs and evaluated on a held-out synthetic test set. As detailed in Table 4, both models demonstrated strong performance, with the NPD Trait model achieving an F1-Score of 0.86 and the Abuse Cycle model achieving an F1-Score of 0.83. The training dynamics, shown in Fig. 7, Fig. 8, Fig. 9 and Fig. 10, confirm successful and stable convergence, evidenced by consistently decreasing loss and increasing accuracy.

Algorithm 1 Synergistic Multi-Label Classification Algorithm

```
Input: Input Sentence
Output: Predicted NPD Traits (a set), Predicted Abuse Stages
        (a set), NPD Method, Abuse Method, NPD
        Confidences (a list)
1. Regex Matching:;
    regex_trait_match ← Check for NPD trait Regex pattern;
    regex_stage_match ← Check for abuse stage Regex
    pattern;
2. NPD Trait Classification:;
    if regex_trait_match then
        Predicted NPD Traits ← {Regex Label};
        // Output is a set NPD Method ← "Regex";
        NPD Confidences ← [1.0];
        // Output is a list of confidences
    end
    else
        // Use the domain-adapted model for prediction
        trait_probabilities ←
        DomainAdaptedBERT(npd_model, Input
        Sentence);
        Predicted NPD Traits ← {};
        // Initialise an empty set NPD Confidences ← [];
        // Initialise an empty list
        // Multi-label logic using a threshold foreach label i
        with probability p_i in trait_probabilities do
            if p_i > 0.5 then
                add label i to Predicted NPD Traits;
                add p_i to NPD Confidences;
            end
        end
        NPD Method ← "NLP";
    end
3. Abuse Stage Classification:;
    if regex_stage_match then
        Predicted Abuse Stages ← {Regex Label};
        // Output is a set Abuse Method ← "Regex";
    end
    else
        // Multi-label logic for abuse stages (though less
        common to have multiple) stage_probabilities ←
        DomainAdaptedBERT(abuse_model, Input
        Sentence);
        Predicted Abuse Stages ← {};
        // Initialise an empty set
        foreach label j with probability p_j in
        stage_probabilities do
            if p_j > 0.5 then
                add label j to Predicted Abuse Stages;
            end
        end
        Abuse Method ← "NLP";
    end
4. Output:;
    Return Predicted NPD Traits, Predicted Abuse Stages,
    NPD Method, Abuse Method, NPD Confidences;
```

These results validate the model architecture's ability to effectively learn the linguistic patterns within the synthetic corpus, providing a robust baseline for the subsequent real-world validation stages.

TABLE 4. Stage 1: Baseline model performance on the synthetic test set. (Source: Authors' own elaboration).

NPD Trait Model	Performance	Abuse Cycle Model	Performance
Accuracy (%)	87.3	**Accuracy (%)**	84.6
Precision	0.85	**Precision**	0.82
Recall	0.87	**Recall**	0.85
F1-Score	0.86	**F1-Score**	0.83

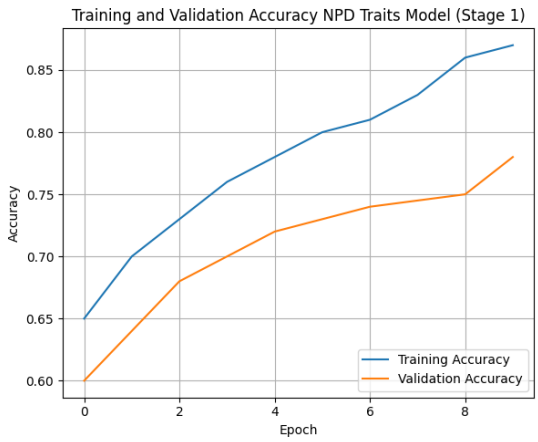

FIGURE 7. Training and validation accuracy across epochs for NPD traits model (Stage 1). (Source: Authors' own elaboration).

B. STAGE 2. BASELINE MODEL PERFORMANCE ON REAL-WORLD DATASETS

To quantify the synthetic-to-real domain gap, the baseline models were evaluated against three distinct real-world datasets. The results, detailed in Table 5, reveal the performance degradation when moving from a synthetic to a real-world context. This performance drop is an empirical quantification of the domain gap. The F1-Score for the NPD Trait model, for instance, fell from 0.86 on synthetic data to as low as 0.52 on the PANDORA corpus, which is characterised by high linguistic variance and contextual ambiguity. The model performed best on the Jigsaw dataset, which shares a thematic focus on toxicity with our DAPT pre-training data, yet still demonstrated a significant performance gap. This multi-faceted analysis underscores the necessity of domain adaptation, as models trained exclusively on synthetic data are insufficiently robust to handle the lexical diversity and stylistic drift inherent in authentic online communication.

C. STAGE 3: PERFORMANCE RESTORATION VIA DOMAIN ADAPTATION ON REAL-WORLD DATASETS

The final evaluation stage was designed to demonstrate the efficacy of Domain-Adaptive Pre-training (DAPT) in mitigating the performance degradation identified in Stage 2. As presented in Table 6, the final, domain-adapted

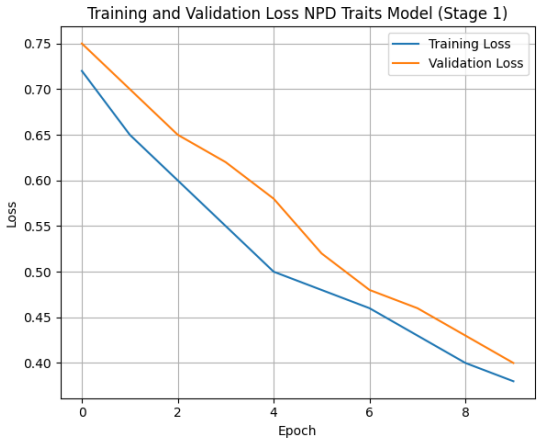

FIGURE 8. Training and validation loss across epochs for NPD traits model (Stage 1). (Source: Authors' own elaboration).

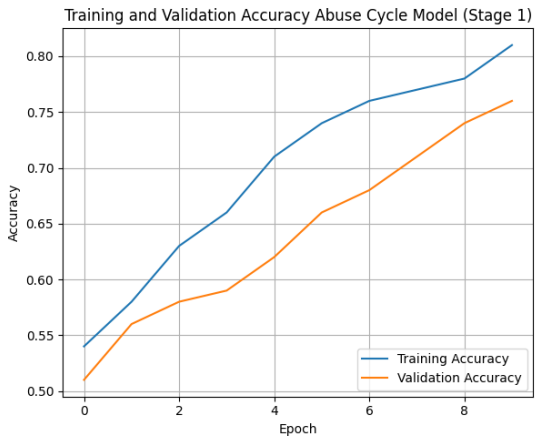

FIGURE 9. Training and validation accuracy across epochs for abuse cycle model (Stage 1). (Source: Authors' own elaboration).

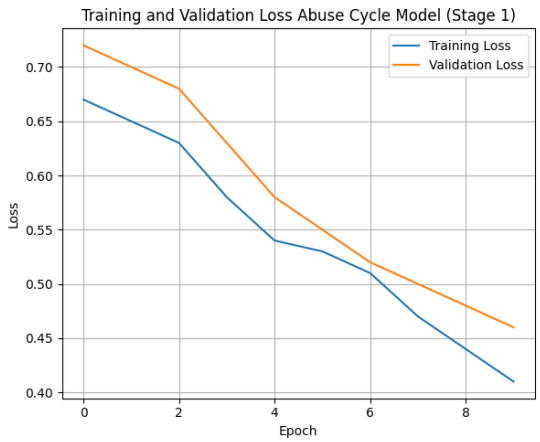

FIGURE 10. Training and validation loss across epochs for abuse cycle model (Stage 1). (Source: Authors' own elaboration).

models achieved a substantial and consistent restoration of performance across all real-world datasets. Notably, the F1-Score for the NPD Trait model on the challenging Jigsaw dataset was restored from 0.65 to 0.82, with significant

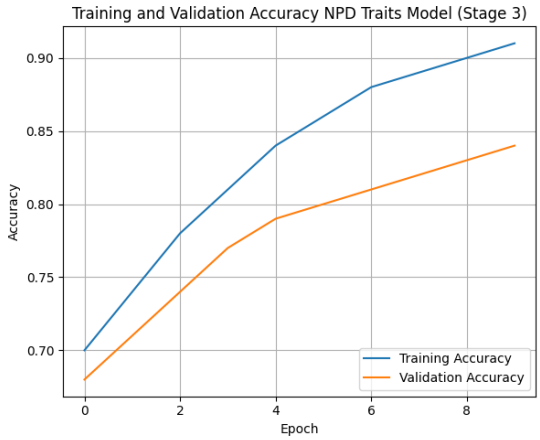

FIGURE 11. Training and validation accuracy across epochs for NPD traits model (Stage 3). (Source: Authors' own elaboration).

improvements also observed on the MBTI and PANDORA datasets (approximately 45% and 52%, respectively).

The fine-tuning process over 10 epochs was stable and successful, demonstrating effective learning and generalisation. This is evidenced by the consistent upward trend in training and validation accuracy (Fig. 11 and Fig. 13) and the corresponding steady decline in loss (Fig. 12 and Fig. 14). For reproducibility, our experiments were conducted using a bert-base-uncased model optimised with AdamW (batch size of 16), with DAPT run for 3 epochs and supervised fine-tuning for 10 epochs.

This marked improvement provides compelling quantitative evidence that the DAPT strategy successfully equipped the models with the necessary linguistic knowledge to generalise from a synthetic training environment to complex, real-world text.

D. ABLATION STUDY FOR ARCHITECTURAL SYNERGY

To validate that the hybrid architecture provides a genuine synergistic benefit, a final analysis was conducted. An ablation study was performed by averaging the performance metrics across the three real-world validation datasets (MBTI, PANDORA and Jigsaw) to provide a comprehensive measure of each component's contribution.

As shown in Table 7, the proposed hybrid model achieved the highest F1-Score (0.82). The Regex-Only model demonstrated high precision (0.92) but suffered from very low recall (0.45), making it unsuitable for comprehensive detection. The BERT-Only model was more balanced but was outperformed by the final hybrid model, which improved precision without sacrificing recall.

To analyse the impact of the domain adaptation corpus size on model performance, an ablation study was conducted. The final model was trained using varying percentages of the "Hate Speech Dataset [35]" DAPT corpus. As shown in Table 8, performance improves significantly as more real-world adaptation data is used, validating the robustness of the DAPT strategy. This result provides quantitative,

TABLE 5. Stage 2 performance of baseline models on multiple real-world datasets. (Source: Authors' own elaboration).

Model	Dataset	F1-Score (Macro Avg.)	Precision (Macro Avg.)	Recall (Macro Avg.)
NPD Trait Model	MBTI Dataset	0.55	0.59	0.52
	PANDORA Dataset	0.52	0.58	0.48
	Jigsaw Dataset	0.65	0.68	0.63
Abuse Cycle Model	MBTI Dataset	0.51	0.55	0.48
	PANDORA Dataset	0.48	0.54	0.44
	Jigsaw Dataset	0.62	0.65	0.60

TABLE 6. Stage 3 Performance of domain-adapted models on real-world datasets. (Source: Authors' own elaboration).

Model	Dataset	F1-Score (Macro Avg.)	Precision (Macro Avg.)	Recall (Macro Avg.)
NPD Trait Model	MBTI Dataset	0.80	0.81	0.79
	PANDORA Dataset	0.79	0.80	0.78
	Jigsaw Dataset	0.82	0.83	0.81
Abuse Cycle Model	MBTI Dataset	0.76	0.77	0.75
	PANDORA Dataset	0.75	0.76	0.74
	Jigsaw Dataset	0.79	0.80	0.78

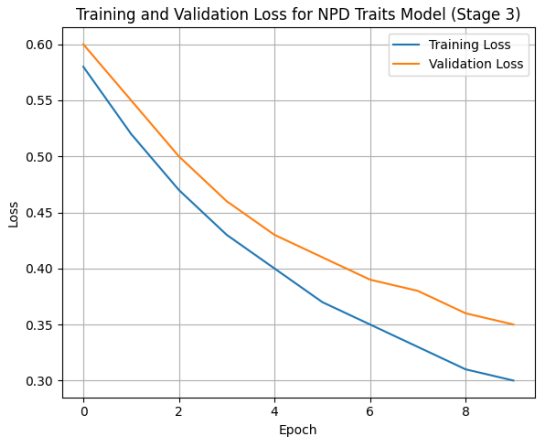

FIGURE 12. Training and validation loss across epochs for NPD traits model (Stage 3). (Source: Authors' own elaboration).

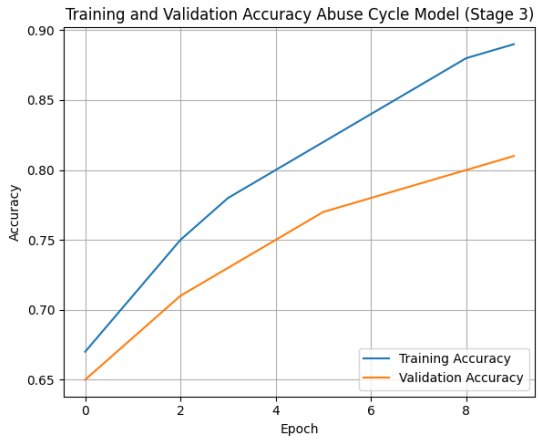

FIGURE 13. Training and validation accuracy across epochs for abuse cycle model (Stage 3). (Source: Authors' own elaboration).

empirical evidence that the integration of both components provides a more balanced and robust classification than either technique could achieve alone, confirming that the architecture is genuinely synergistic.

E. QUALITATIVE AND COMPARATIVE ANALYSIS

To demonstrate the model's practical efficacy, we performed a qualitative analysis of its components and a quantitative comparison against a general-purpose Large Language Model (LLM, ChatGPT, GPT-40). The system's complementary strengths are highlighted below:

- Regex Precision: The high-precision regex component accurately classifies unambiguous, explicit phrases (e.g., "You are a worthless piece of trash"), reliably detecting overt markers.
- NLP Contextual Understanding: The domain-adapted BERT model correctly interprets implicit and sarcastic text (e.g., "Wow, another brilliant idea from the genius over here"), a task impossible for static rules.
- Hybrid Synergy: The combined architecture ensures reliable detection of explicit markers via regex, with the NLP model providing a fallback for nuanced expressions. The system's outputs are interpretable, clearly indicating which component made the classification.
- Error Analysis: A key limitation was identified in contextual irony; a phrase like "…ironically call each other 'worthless' as a joke" was misclassified as `Verbal Abuse` because the high-priority regex component superseded the NLP model's contextual analysis.

The system also demonstrated its utility for analysing longer narratives, such as chat logs, by iteratively processing sentence segments. For a comparative benchmark, we evaluated ChatGPT (GPT-4o) on 50 sentences. The model

TABLE 7. Ablation study of model components on the real-world dataset. (Source: Authors' own elaboration).

Model Architecture	F1-Score (Macro Avg.)	Precision (Macro Avg.)	Recall (Macro Avg.)
Regex-Only	0.60	0.92	0.45
BERT-Only (Domain-Adapted)	0.80	0.80	0.81
Hybrid Model (Proposed)	0.82	0.83	0.81

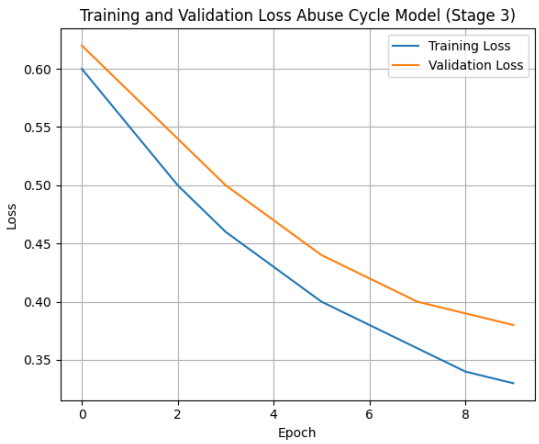

FIGURE 14. Training and validation loss across epochs for abuse cycle model (Stage 3). (Source: Authors' own elaboration).

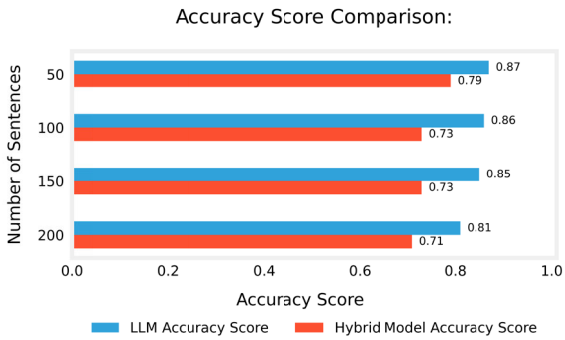

FIGURE 15. Aggregate accuracy comparison with LLM (GPT-4o). (Source: Authors' own elaboration).

was provided with our taxonomy of 24 NPD traits and 4 abuse cycle stages and prompted to classify each sentence accordingly. For each classification, ChatGPT was also instructed to provide a confidence score for both the assigned NPD trait and the abuse cycle stage (Fig. 16). Furthermore, an aggregate accuracy comparison demonstrates that while the LLM performs strongly, our model maintains more consistent performance as the volume of text increases, highlighting its robustness for scalable analysis in this domain (Fig. 15).

V. DISCUSSION

The successful development and rigorous multi stage validation of the hybrid classification system have yielded several key findings. This section interprets the empirical results, focusing on the successful mitigation of the synthetic-to-real domain gap and the evidence for the model's architectural synergy. Furthermore, it addresses the significant ethical considerations inherent in this work by proposing concrete mitigation strategies and concludes by discussing the study's limitations and promising directions for future research.

A. STRENGTHS AND NOVELTY

The primary contribution is a robust methodological framework for developing a specialised classifier in a data-scarce psychological domain. The key strengths are:

1) A Novel End-to-End Validation Framework: This study introduces a complete framework to bridge the synthetic-to-real domain gap. It combines synthetic data generation, unsupervised Domain-Adaptive Pre-training (DAPT) and a rigorous three stage evaluation to create a replicable blueprint for real-world applications. The framework's success is demonstrated in our multi stage evaluation (Table 6 and Table 7). Efficacy on informal text is ensured by augmenting the regex library with over 200 curated slang terms per category.
2) Empirically Proven Architectural Synergy:A formal ablation study provides empirical evidence of architectural synergy. Results show the final hybrid model achieves a higher F1-score than either its rule-based regex or domain-adapted BERT components in isolation, confirming the components work together to create a more robust and balanced system.
3) Enhanced Psychological Validity: Our multi-label classification framework enhances psychological validity over single-label approaches. It allows the model to predict multiple co-occurring NPD traits within a single utterance, which more accurately reflects the complex nature of narcissistic communication.

B. ETHICAL IMPLICATIONS AND RESPONSIBLE AI

The development of AI tools for detecting personality disorders, such as NPD, carries significant ethical implications that warrant careful and ongoing consideration as mentioned in Table 9. While our research aims to provide a computational tool for analysing textual data for patterns relevant to narcissistic dynamics, potentially aiding researchers and clinicians, the potential for misuse and negative societal impact must be proactively addressed.

To move beyond identifying these risks, we propose a framework of actionable mitigation strategies:

TABLE 8. DAPT ablation study. (Source: Authors' own elaboration).

% of DAPT Corpus Used	F1-Score (Macro)	Precision (Macro)	Recall (Macro)
0% (Baseline)	0.65	0.68	0.63
10%	0.68	0.71	0.66
50%	0.75	0.77	0.74
100% (Full Model)	0.82	0.83	0.81

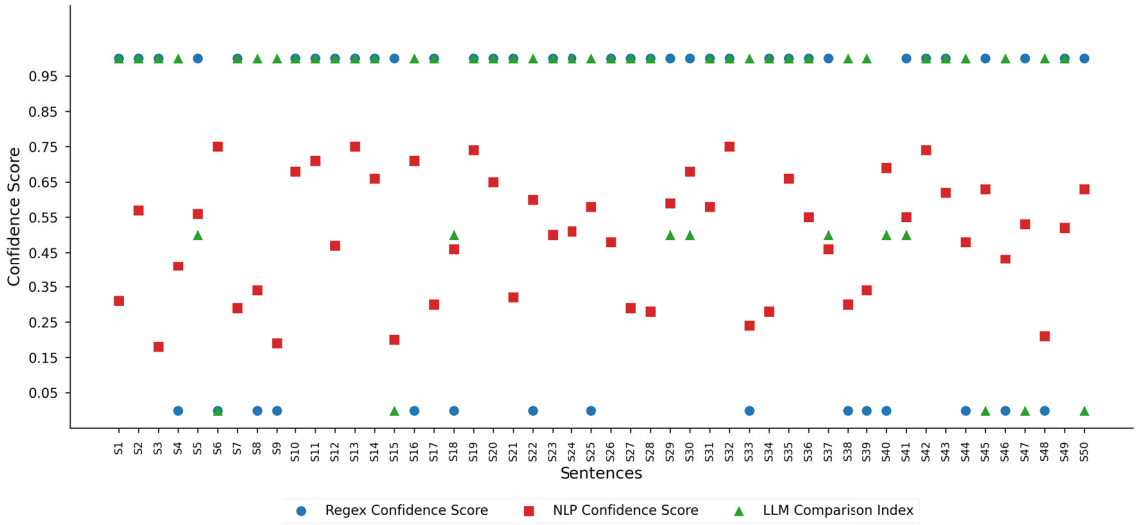

FIGURE 16. Per-Sentence confidence score comparison with LLM (GPT-4o). (Source: Authors' own elaboration).

TABLE 9. Key ethical considerations. (Source: Authors' own elaboration).

No.	Consideration Area	Summary
1.	Risk of Misdiagnosis and Stigmatisation	Danger of mislabelling individuals or pathologising normal communication; tool is not for diagnosis.
2.	Accuracy, Bias and Fairness	Models susceptible to data bias; risk of unfair performance across demographics; requires continuous auditing.
3.	Privacy and Data Security	Handling sensitive text data requires strict compliance (GDPR, HIPAA), anonymisation and security.
4.	Autonomy and Consent	Individuals must be informed and consent to personality-related analysis of their data.
5.	Potential for Misuse	Risks include interpersonal harm, surveillance/profiling, misuse in adversarial contexts and affecting clinical relationships.
6.	Societal Perceptions and "Techno-solutionism"	Risk of oversimplifying complex psychological issues through technology, undermining human judgment.

1) Technical Safeguards: To mitigate bias, the model must be subjected to regular audits using fairness toolkits to ensure equitable performance across demographic groups. For any application handling user data, strict compliance with privacy regulations like GDPR and HIPAA is non-negotiable, enforced through technical measures including end-to-end encryption, data anonymisation and robust role-based access controls.
2) Procedural Safeguards: The system's intended use must be explicitly defined as a support tool for trained researchers and clinicians, not as a diagnostic tool for the public. Any user-facing application built on this technology must include:
   - A clear and unavoidable AI disclosure, informing users they are interacting with an automated system.
   - Strong warnings against self-diagnosis or using the tool to label others.
   - An integrated crisis-response protocol to detect language related to self-harm and immediately provide resources for crisis services, such as suicide prevention hotlines.

Our research is currently focused on computational linguistics and pattern recognition to understand how narcissistic dynamics may manifest in language. Any future development towards real-world application would necessitate a

multidisciplinary approach, involving ethicists, clinicians, legal experts and affected communities.

C. PRACTICAL APPLICATIONS AND RECOMMENDATIONS

The framework developed in this study offers several concrete proposals for practical application, providing measurable benefits in various domains:

- Therapeutic and Clinical Settings: The system can assist therapists by analysing patient communications (e.g., journals) by automatically flagging linguistic patterns of idealisation or devaluation to support clinical judgement and track therapeutic progress.
- Legal and Forensic Contexts: The tool can analyse digital communications to identify patterns of manipulation and coercive control. This offers objective, data-driven evidence to support legal arguments in cases involving psychological abuse.
- Digital Platform Moderation: The framework enables proactive content moderation by detecting targeted harassment and grooming that follow the idealisation-devaluation cycle. This phase-aware approach surpasses keyword filters by identifying malicious intent without explicit slurs, enhancing user safety.

D. LIMITATIONS AND FUTURE DIRECTIONS

While this study successfully demonstrates a robust methodology for creating and validating a specialised classifier, certain limitations must be acknowledged. The primary limitation is that our real-world validation, while extensive, was conducted on proxy datasets of general toxicity rather than a "gold-standard" corpus annotated specifically for NPD by clinical experts. Although our multi-label framework was implemented to address the issue of trait overlap, the inherent ambiguity and fuzzy boundaries between psychological constructs can still pose a challenge. The proxy datasets used for validation (e.g., Jigsaw, PANDORA) are unlikely to contain the nuanced linguistic patterns of the full narcissistic abuse cycle, such as 'idealisation' or 'hoovering'. Therefore, the model's phase-aware classification component remains in need of more validation in a real-world context. Furthermore, the current model analyses sentences in isolation, which limits its ability to capture broader dialogue context, potentially leading to misinterpretation of sarcasm or irony.

Future research should prioritise building upon the successful validation framework established in this work. The most critical next step is the curation and validation of the system against a clinically annotated, real-world dataset (e.g., anonymised therapy transcripts or online support forum discussions) to assess its true clinical utility. Building on our architectural improvements, exploring more sophisticated NLP models that incorporate multi-sentence or dialogue context could further enhance performance and mitigate the context sensitivity limitation. Finally, while this paper proposes a strong ethical framework, future work must focus on its practical implementation and iterative refinement in collaboration with clinicians, ethicists and affected communities before any real-world deployment is considered.

VI. CONCLUSION

This study introduced and validated a novel end-to-end framework for detecting linguistic markers of narcissistic personality disorder. By employing a synergistic hybrid Regex-NLP model within a three-stage validation process, we demonstrated a replicable methodological blueprint that successfully bridges the synthetic-to-real domain gap. The resulting system shows robust performance on real-world proxy data, establishing a strong foundation for developing computational tools for researchers and clinicians. While the next critical step is validation against clinically annotated corpora, this work provides a valuable blueprint for creating and verifying responsible AI tools in computational psychology.

REFERENCES

[1] P. K. Jonason, M. Lyons, E. J. Bethell, and R. Ross, "Different routes to limited empathy in the sexes: Examining the links between the dark triad and empathy," *Personality Individual Differences*, vol. 54, no. 5, pp. 572–576, Apr. 2013.

[2] G. A. Keepers, L. J. Fochtmann, J. M. Anzia, S. Benjamin, J. M. Lyness, R. Mojtabai, M. Servis, L. W. Choi-Kain, K. Nelson, J. M. Oldham, C. Sharp, A. Degenhardt, L. J. Fochtmann, J. M. Oldham, S.-H. Hong, and J. Medicus, "The American psychiatric association practice guideline for the treatment of patients with borderline personality disorder," *Amer. J. Psychiatry*, vol. 181, no. 11, pp. 1024–1028, Nov. 2024.

[3] *Diagnostic and Statistical Manual of Mental Disorders Psychiatry Online*. Accessed: Aug. 12, 2025. [Online]. Available: https://psychiatryonline.org/doi/book/10.1176/appi.books.9780890425596

[4] New Beginnings Therapy. *The Psychology of Narcissism—W*. Accessed: Sep. 30, 2025. [Online]. Available: https://newbeginningstherapy.co.nz/2018/01/21/psychology-narcissism-w-keith-campbell/

[5] M. L. Oster and J. Hines, "Media review: The narcissism epidemic: Living in the age of entitlement," *J. Student Affairs Res. Pract.*, vol. 48, no. 3, pp. 385–387, Jul. 2011, doi: 10.2202/1949-6605.6395.

[6] I. W. Meier, "The dark tetrad and the perpetration of cyber dating abuse," M.S. thesis, Dept. Commun. Cognition, School Humanities Digit. Sci., Tilburg Univ., Tilburg, The Netherlands, 2023. [Online]. Available: https://arno.uvt.nl/show.cgi?fid=161537

[7] A. Cohen, *The Cyber Predators: Dark Personality and Online Misconduct and Crime*. Cambridge, U.K.: Cambridge Univ. Press, Apr. 2024.

[8] P. Mitra, T. J. Torrico, and D. Fluyau, *Narcissistic Personality Disorder*. Treasure Island, FL, USA: StatPearls, 2025.

[9] Choosing Therapy Editorial Team. (2024). *The Narcissistic Abuse Cycle: A Comprehensive Guide*. [Online]. Available: https://www.choosingtherapy.com/narcissistic-abuse-cycle/

[10] S. Bliss. *Navigating the Narcissist's Idealization: Mastering the Art of Gray Rock*. Accessed: Aug. 12, 2025. [Online]. Available: https://medium.com/@suzybliss/navigating-the-narcissists-idealization

[11] D. Lancer. (Mar. 2022). *The Narcissistic Cycle of Abuse*. Accessed: Sep. 30, 2025. [Online]. Available: https://www.psychologytoday.com/gb/blog/love-in-the-age-narcissism/202203/the-narcissistic-cycle-abuse

[12] J. W. Pennebaker, R. L. Boyd, K. Jordan, and K. Blackburn, "The development and psychometric properties of LIWC2015," Univ. Texas at Austin, Austin, TX, USA, Tech. Rep., Sep. 2015.

[13] G. Park, H. A. Schwartz, J. C. Eichstaedt, M. L. Kern, M. Kosiński, D. Stillwell, L. Ungar, and M. E. P. Seligman, "Automatic personality assessment through social media language," *J. Personality Social Psychol.*, vol. 108, no. 6, pp. 934–952, Jun. 2014.

[14] K. Safjan. (2023). *From Fixed-Size to NLP Chunking—A Deep Dive Into Text Chunking Techniques*. [Online]. Available: https://www.safjan.com/from-fixed-size-to-nlp-chunking-a-deep-dive-into-text-chunking-techniques/

[15] R. Chatila and J. C. Havens, "The IEEE global initiative on ethics of autonomous and intelligent systems," in *Robotics and Well-Being*, vol. 95, M. I. A. Ferreira, J. S. Sequeira, G. S. Virk, M. O. Tokhi, and E. E. Kadar, Eds., Cham, Switzerland: Springer, 2019, pp. 11–16.

[16] C. Sekhar, M. S. Rao, A. S. K. Nayani, and D. Bhattacharyya, "Emotion recognition through human conversation using machine learning techniques," in *Machine Intelligence and Soft Computing*, D. Bhattacharyya and N. T. Rao, Eds., Singapore: Springer, 2021, pp. 113–122.
[17] T. Al Hanai, M. Ghassemi, and J. Glass, "Detecting depression with audio/text sequence modeling of interviews," in *Proc. Interspeech*, Sep. 2018, pp. 1716–1720.
[18] A. Alslaity and R. Orji, "Machine learning techniques for emotion detection and sentiment analysis: Current state, challenges, and future directions," *Behaviour Inf. Technol.*, vol. 43, no. 1, pp. 139–164, Jan. 2024.
[19] A. Mitra, A. Biswas, K. Chakraborty, A. Ghosh, N. Das, N. Ghosh, and A. Ghosh, "A machine learning approach to identify personality traits from social media," in *Machine Learning and Deep Learning in Efficacy Improvement of Healthcare Systems*. Abingdon, U.K.: CRC Press, 2022, pp. 31–59.
[20] T. Zhang, A. M. Schoene, S. Ji, and S. Ananiadou, "Natural language processing applied to mental illness detection: A narrative review," *Npj Digit. Med.*, vol. 5, no. 1, pp. 1–13, Apr. 2022.
[21] M. Jayaratne and B. Jayatilleke, "Predicting personality using answers to open-ended interview questions," *IEEE Access*, vol. 8, pp. 115345–115355, 2020.
[22] J. Jang, S. Yoon, G. Son, M. Kang, J. Y. Choeh, and K.-H. Choi, "Predicting personality and psychological distress using natural language processing: A study protocol," *Frontiers Psychol.*, vol. 13, Apr. 2022, Art. no. 865541.
[23] H. A. Alatabi and A. R. Abbas, "Sentiment analysis in social media using machine learning techniques," *Iraqi J. Sci.*, vol. 61, pp. 193–201, Jan. 2020.
[24] Z. Wang, Y. Liu, J. Fang, and D. Li, "Deep learning-based sentiment analysis for social media," in *Proc. 5th Int. Conf. Artif. Intell. Pattern Recognit.* New York, NY, USA: ACM, May 2022, pp. 30–37.
[25] J. Pandey and S. V. A. Rao, "A survey on sentiment analysis and opinion mining," in *Proc. Int. Conf. Adv. Inf. Commun. Technol. Comput.* New York, NY, USA: ACM, Aug. 2016, pp. 1–5.
[26] AI BY TEC. *The Future of AI in Detecting Narcissistic Personality Disorder(NPD)*. Accessed: Sep. 30, 2025. [Online]. Available: https://www.linkedin.com/pulse/future-ai-detecting-narcissistic-personality-ai-by-tec-yfoif/
[27] J. Jia, W. Liang, and Y. Liang, "A review of hybrid and ensemble in deep learning for natural language processing," 2023, *arXiv:2312.05589*.
[28] J. Villena, S. Collada-Pérez, S. Serrano, and J. Gonzalez-Cristobal, "Hybrid approach combining machine learning and a rule-based expert system for text categorization," in *Proc. 24th Int. Florida Artif. Intell. Res. Soc. Conf.*, Jan. 2011, pp. 1–6.
[29] (Sep. 2020). *The Hybrid Approach To Enhancing Your Natural Language Processing*. [Online]. Available: https://blogs.sas.com/content/subconsciousmusings/
[30] A. Le Glaz, Y. Haralambous, D.-H. Kim-Dufor, P. Lenca, R. Billot, T. C. Ryan, J. Marsh, J. DeVylder, M. Walter, S. Berrouiguet, and C. Lemey, "Machine learning and natural language processing in mental health: Systematic review," *J. Med. Internet Res.*, vol. 23, no. 5, May 2021, Art. no. e15708.
[31] J. Ive, N. Viani, J. Kam, L. Yin, S. Verma, S. Puntis, R. N. Cardinal, A. Roberts, R. Stewart, and S. Velupillai, "Generation and evaluation of artificial mental health records for natural language processing," *Npj Digit. Med.*, vol. 3, no. 1, pp. 1–9, May 2020.
[32] J. Ive, "Leveraging the potential of synthetic text for AI in mental healthcare," *Frontiers Digit. Health*, vol. 4, pp. 1–5, Oct. 2022. [Online]. Available: https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2022.1010202/full
[33] C. Nobata, J. Tetreault, A. Thomas, Y. Mehdad, and Y. Chang, "Abusive language detection in online user content," in *Proc. 25th Int. Conf. World Wide Web*, Apr. 2016, pp. 145–153.
[34] *Personality Disorder—PS3018—Clinical Psychology Lecture 9 Personality Disorders Personality—Studocu*. Accessed: Aug. 12, 2025. [Online]. Available: https://www.studocu.com/en-gb/document/city-university-of-london/clinical-psychology/personality-disorder/15284555
[35] T. Davidson, D. Warmsley, M. Macy, and I. Weber. (2017). *Hate Speech and Offensive Language Dataset*. Accessed: Sep. 30, 2025. [Online]. Available: https://www.kaggle.com/datasets/mrmorj/hate-speech-and-offensive-language-dataset
[36] *(MBTI) Myers-Briggs Personality Type Dataset*. Accessed: Sep. 24, 2025. [Online]. Available: https://www.kaggle.com/datasets/datasnaek/mbti-type
[37] M. Gjurković, M. Karan, I. Vukojević, M. Bošnjak, and J. Šnajder, "PANDORA talks: Personality and demographics on Reddit," Univ. Zagreb, Zagreb, Croatia, Tech. Rep., Jun. 2021.
[38] *Jigsaw Toxic Comment Classification Challenge*. Accessed: Sep. 24, 2025. [Online]. Available: https://www.kaggle.com/competitions/jigsaw-toxic-comment-classification-challenge/data
[39] *Administrative Data Research U.K. Data-Driven Change—ADR U.K.* Accessed: Sep. 24, 2025. [Online]. Available: https://www.adruk.org/
[40] *Harmony—A Global Platform for Contextual Data Harmonisation*. Accessed: Sep. 24, 2025. [Online]. Available: https://harmonydata.ac.uk/

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